

Lectures for the School on Quantum Simulation

Alessandro Roggero



Milan
– 9-11 September 2026 –



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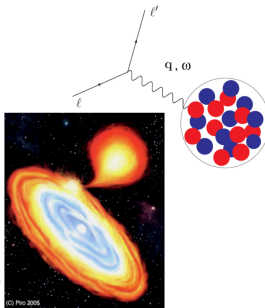
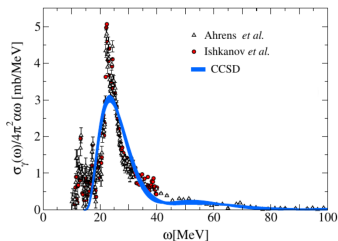
	9 Sept	10 Sept	11 Sept
8:30-9:00	Registration		
9:00-9:30	Welcome+Sponsors	THEORY 3 VQE, intro to errors and their mitigation	THEORY 5 ancilla-free spectral function, circuit knitting
9:30-10:30	THEORY 1 Quantum Simulations		
10:30-11:00	Break	Break	Break
11:00-12:30	EXERCISE 1 Time evolution for Ising chain	EXERCISE 3 VQE and read-out error mitigation	THEORY 6 advanced error mitigation, LCU and QSP
12:30-14:00	Lunch	Lunch	Lunch
14:00-14:20	THEORY 2 Trotter-Suzuki decomposition, linear response	LINKS Foundation	EXERCISE 5 full protocol implementation on hardware
14:20-15:30		THEORY 4 error mitigation for quantum gates	
15:30-16:00	Break	Break	Break
16:00-17:30	EXERCISE 2 Correlation functions and execution on emulators	EXERCISE 4 Execution on quantum hardware with full mitigation	EXERCISE 6 full protocol implementation on hardware, extensions

Preparing low energy states: why?

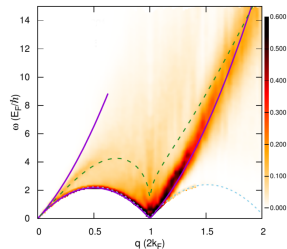
General scheme for many-body quantum simulations

- Discretize physical problem on finite Hilbert space
- Encode discrete problem into spin problem
- Prepare an encoded low energy state
- Manipulate state, e.g. evolve under unitary time evolution
- Measure properties of final state

Bacca et al. PRL(2013)



Motta et al. PRA(2016)



Preparing low energy states: how?

Ground state preparation is not easy!

For a general Hamiltonian H , preparing the ground-state $|\Psi_0\rangle$ or even just estimating its energy E_0 is beyond what QC can do efficiently (QMA-hard)

Physically interesting Hamiltonians are not general so we might be lucky.

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Lots of empirically successful classical methods for this, can we use them?

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Methods that implement energy filters to boost ground-state

- classical: Lanczos/Krylov iterations, Imaginary time evolution (QMC)

$$|\Phi(\tau)\rangle \propto \exp(-\tau H) |\Phi(0)\rangle \rightarrow |\Psi_0\rangle + \exp(-\tau(E_1 - E_0)) |\Psi_1\rangle + \dots$$

- quantum: we can do something similar but
 - need to implement non-unitary evolution (see Friday class)
 - need to start with a decent state $|\Phi(0)\rangle$ to be efficient

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Methods that implement energy filters to boost ground-state

Methods that use the variational principle

Take some parametrized state $|\phi(\vec{\alpha})\rangle$ for which we can measure the energy expectation value $E(\vec{p}) = \langle \phi(\vec{\alpha}) | H | \phi(\vec{\alpha}) \rangle \geq E_0$. But then

$$E(\vec{\alpha}_{best}) = \min_{\vec{\alpha}} E(\vec{\alpha}) \quad \rightarrow \quad |\phi(\vec{\alpha}_{best})\rangle \approx |\Psi_0\rangle$$

Accuracy not guaranteed but can use physical intuition in parametrization

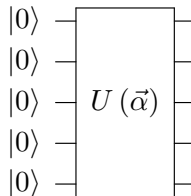
Variational Quantum Eigensolver (VQE)

see e.g. Perruzzo et al. (2014), McClean et.al. (2016), Cerezo, Arrasmith, et al. (2021)

Variational State Preparation

Exploit variational principle for the energy to find some reasonable parametrization for the ground-state

$$E(\vec{\alpha}) = \langle \Psi(\vec{\alpha}) | H | \Psi(\vec{\alpha}) \rangle \geq E_0$$



- On a quantum computer we use a unitary ansatz $|\Psi(\vec{\alpha})\rangle = U(\vec{\alpha})|0\rangle$

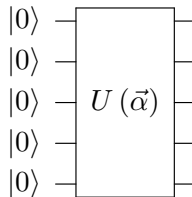
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- On a quantum computer we use a unitary ansatz $|\Psi(\vec{\alpha})\rangle = U(\vec{\alpha})|0\rangle$
- To get the energy we can first decompose into Pauli operators

$$H = \sum_{l=1}^L c_l P_l \quad \rightarrow \quad E(\vec{\alpha}) = \sum_{l=1}^L c_l \langle \Psi(\vec{\alpha}) | P_l | \Psi(\vec{\alpha}) \rangle$$

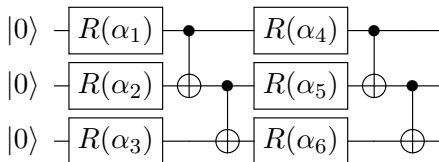
- Use some classical optimization algorithm to find optimum

Since we are forced to evaluate expectation values in groups of commuting operators, the variance might be large even if we have exact state!

Folklore about VQE and Near Intermediate Scale Quantum

VQE is NISQ friendly since you get the most out of what you have

I have a QPU with only 3 qubits and can do at most 4 CNOT gates



Variational methods usually work well when we use wisely a **physically informed ansatz**, not when we employ one chosen only to be **hardware efficient**

Many VQE variations with physically motivated circuits (UCC, ADAPT, ...)

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VQE is NISQ friendly since you leverage classical HPC for optimization

Non-convex optimization is one of the most challenging computational task there are. Modern neural networks offset some of the cost using automatic differentiation: cheap and accurate gradients of cost function

- with VQE, gradients can have substantial statistical noise
- reasonably simple circuits are hard to simulate classically
- for these circuits the gradients are (often) vanishingly small

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VQE is NISQ friendly since it is resilient against hardware errors

A simple model for a **coherent error** is over-rotation. We want

$$R(\theta) = e^{-i\theta Z}$$

but the machine gives us

$$R_N(\theta) = e^{-i(\theta+\delta)Z} \equiv R(\theta + \delta)$$

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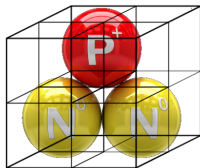
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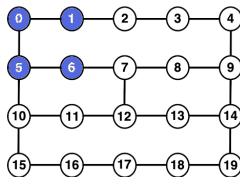
When we optimize from the results obtained on hardware our circuit is actually using R_N . But this is not a problem since we will simply find a different value for the optimal parameters after optimization

- not necessarily as nice as it sounds

How VQE was back in 2019

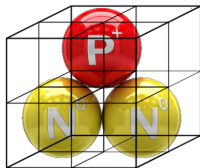


AR, Li, Carlson et al. PRD(2020)

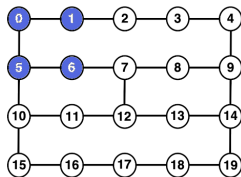


Extremely simplified model for the triton nucleus on 16 states (4 qubits)

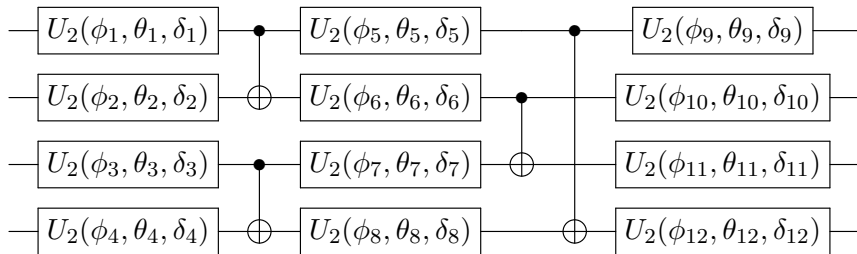
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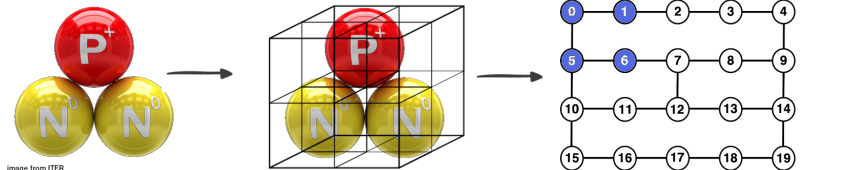


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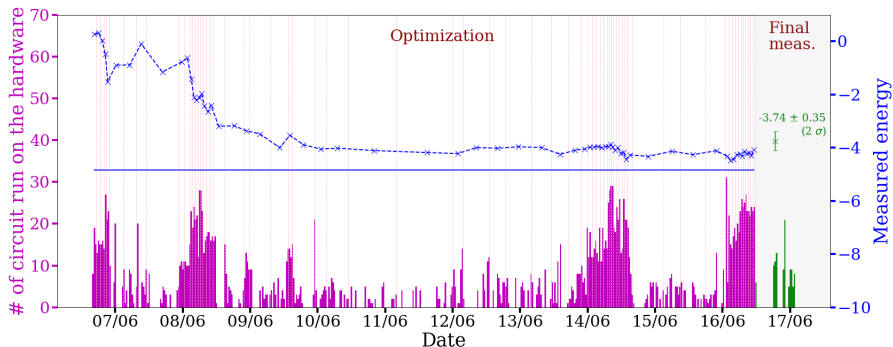


- up to 36 parameters → could be reduced using symmetries

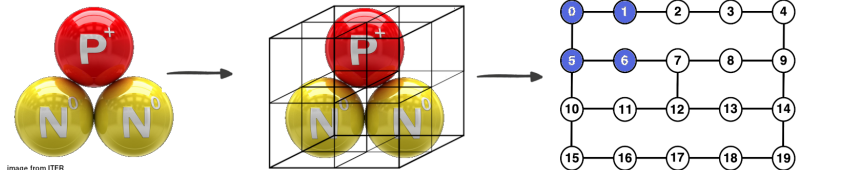
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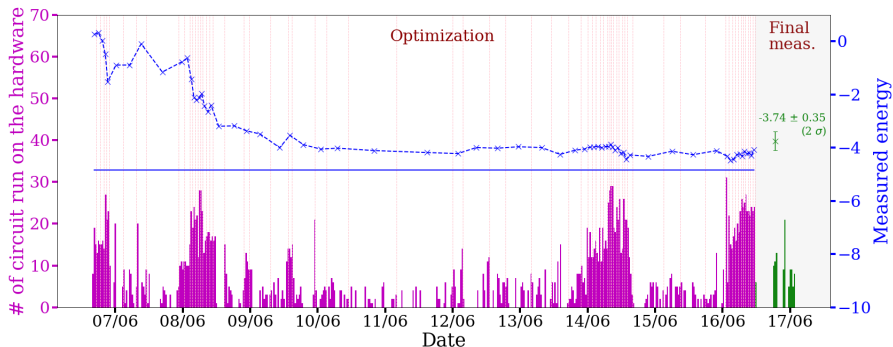
- first problem: latency



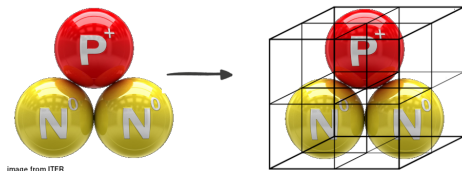
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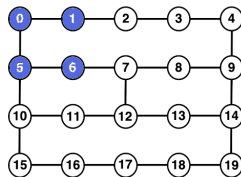
- first problem: latency $\rightarrow$ more compact trial states, better optimizers



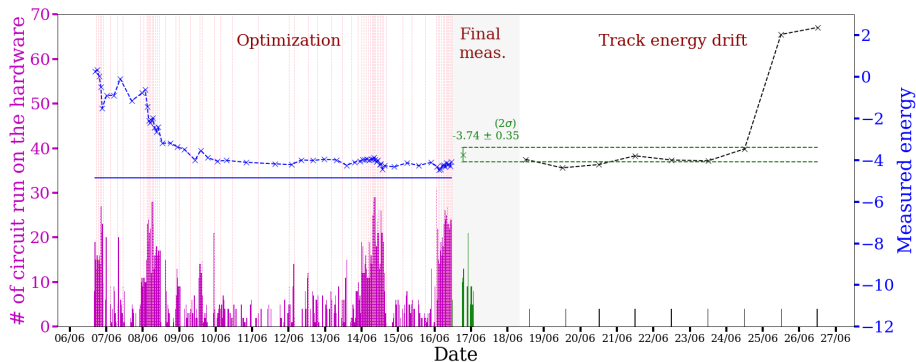
How VQE was back in 2019



AR, Li, Carlson et al. PRD(2020)



- second problem: persistence



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VQE is NISQ friendly and therefore Quantum Machine Learning is too

VQE ansatz for our Ising chain

What ansatz should we use to get a decent low energy state?

Mario's idea: take inspiration from 2 site system!

$$H = -Z_0 - Z_1 + JX_0X_1 = \begin{pmatrix} -2 & 0 & 0 & J \\ 0 & 0 & J & 0 \\ 0 & J & 0 & 0 \\ J & 0 & 0 & 2 \end{pmatrix}$$

- when the two spins are not aligned we have the two eigenstates

$$|\psi_{\pm}\rangle = \frac{1}{\sqrt{2}} (|01\rangle \pm |10\rangle) \quad E_{\pm} = \pm J$$

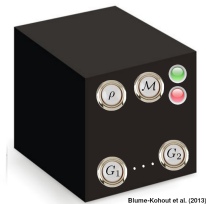
- when the spins are aligned we have eigenvalues $\lambda_{\pm} = \pm\sqrt{J^2 + 4}$ so

$$|\Psi(\theta)\rangle = \cos(\theta) |00\rangle + \sin(\theta) |11\rangle = \exp(-i\theta XY) |00\rangle$$

We'll use this rotation also for larger chains applying it to neighbors. This should be good for $J \ll 1$, at large J we use this as initial perturbed state

First steps towards noise mitigation

we pretended to have this



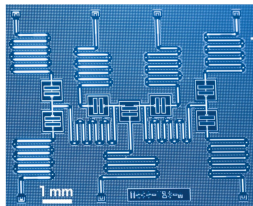
Blume-Kohout et al. (2013)

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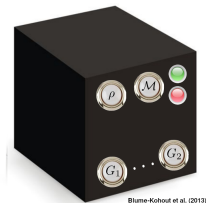


but in reality we have this

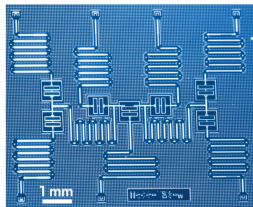


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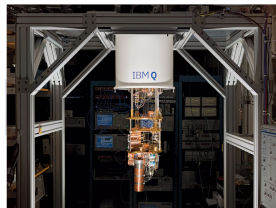
we pretended to have this



but in reality we have this



...inside of this

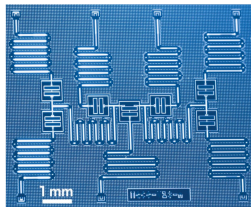


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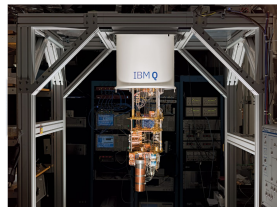
we pretended to have this



but in reality we have this



...inside of this



Errors can broadly be distinguished into two main types

COHERENT we really wanted to use unitary U but we applied unitary U_N

INCOHERENT interaction with the environment generates mixed states

$$\rho = |\psi\rangle\langle\psi| \rightarrow \rho \otimes \rho_{\text{fridge}} \rightarrow \rho_N = \text{Tr}_{\text{fridge}} [\Lambda(\rho \otimes \rho_{\text{fridge}})]$$

After joint evolution system and fridge get entangled, as a result we have

$$\rho_N = (1 - p)\rho + p\rho_{\text{garbage}} \rightarrow \text{Tr}[\rho^2] = 1, \text{Tr}[\rho_N^2] < 1$$

The environment can be your friend

What are we actually doing when we **measure** a quantum system?



$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle \leftrightarrow \rho = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}$$

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$$|\Phi_M\rangle = U_{measure}(|\Psi\rangle \otimes |meter\rangle) = \alpha|00\rangle + \beta|11\rangle$$

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- the outcomes are stored in the (reduced) state of the meter

$$\begin{aligned} \rho_{meter} &= \text{Tr}_{sys}[\rho_{measure}] \equiv \text{Tr}[|0\rangle\langle 0| \otimes \mathbb{1} \rho_{measure}] + \text{Tr}[|1\rangle\langle 1| \otimes \mathbb{1} \rho_{measure}] \\ &= |\alpha|^2|0\rangle\langle 0| + |\beta|^2|1\rangle\langle 1| = \begin{pmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{pmatrix} \end{aligned}$$

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The environment can be your friend, **but don't trust it**

What are we actually doing when we **measure** a quantum system?



$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle \leftrightarrow \rho = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}$$

In an ideal measurement the system density matrix contains the bit probabilities $p_0 = |\alpha|^2$ and $p_1 = |\beta|^2 = 1 - p_0$ on the diagonal

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The simplest model for a measurement error is a random bit-flip

$$\rho_{err} = \mathcal{E}_{bf}[\rho] = (1 - p_e)\rho + p_e X \rho X$$

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Using this simple model, the measured outcomes are sampled instead from

$$\mathcal{E}_{bf}[\rho_{sys}] = p_0^{(e)}|0\rangle\langle 0| + p_1^{(e)}|1\rangle\langle 1| \quad \begin{pmatrix} p_0^{(e)} \\ p_1^{(e)} \end{pmatrix} = \begin{pmatrix} 1 - p_e & p_e \\ p_e & 1 - p_e \end{pmatrix} \begin{pmatrix} p_0 \\ p_1 \end{pmatrix}$$

Notebook time I

We can now take a look at “ErrorMitigation-ReadOut.ipynb”

A bit more on error models

Bit-flip error channel

$$\rho_{err} = \mathcal{E}_{bf} [\rho] = (1 - p_e)\rho + p_e X \rho X$$

We can write a single qubit density matrix in the Pauli basis as follows

$$\rho = \frac{1}{2} \left(\mathbb{1} + \sum_k \text{Tr}[\sigma_k \rho] \sigma_k \right) = \frac{1}{2} (\mathbb{1} + \langle \vec{\sigma} \rangle \cdot \vec{\sigma})$$

The Pauli matrices anticommute with each other meaning that

$$\{\sigma_j, \sigma_k\} \equiv \sigma_j \sigma_k + \sigma_k \sigma_j = \delta_{jk}$$

But then under a bit-flip channel we have that

$$\begin{aligned} \mathcal{E}_{bf} [\rho] &= \left(\frac{1 - p_e}{2} \mathbb{1} + \frac{p_e}{2} \mathbb{1} \right) + \left(\frac{1 - p_e}{2} \langle X \rangle X + \frac{p_e}{2} \langle X \rangle X \right) \\ &\quad + \left(\frac{1 - p_e}{2} \langle Y \rangle Y - \frac{p_e}{2} \langle Y \rangle Y \right) + \left(\frac{1 - p_e}{2} \langle Z \rangle Z - \frac{p_e}{2} \langle Z \rangle Z \right) \end{aligned}$$

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$$\mathcal{E}_{bf}[\rho] = \frac{1}{2} (\mathbb{1} + \langle X \rangle X + (1 - 2p_e) \langle Y \rangle Y + (1 - 2p_e) \langle Z \rangle Z)$$

$$\mathcal{E}_{bf}^R[\rho] = \frac{1}{2} (\mathbb{1} + \langle X \rangle X + (1 - 2p_e)^R \langle Y \rangle Y + (1 - 2p_e)^R \langle Z \rangle Z)$$

Repeated applications remove components that don't commute with X

A bit more on error models: stochastic Pauli channels

Bit-flip (Pauli X) error channel

$$\rho_x = \mathcal{E}_{bf}[\rho] = (1 - p_X)\rho + p_X X\rho X \rightarrow \frac{1}{2}(\mathbb{1} + \langle X \rangle X)$$

Phase-flip (Pauli Z) error channel

$$\rho_z = \mathcal{E}_{pf}[\rho] = (1 - p_Z)\rho + p_Z Z\rho Z \rightarrow \frac{1}{2}(\mathbb{1} + \langle Z \rangle Z)$$

- this is also called **dephasing**, an ideal measurement does just this
- important physical mechanism in many devices $T_2 \approx -\log(p_Z)$

Bit/phase-flip (Pauli Y) error channel

$$\rho_y = \mathcal{E}_{bpf}[\rho] = (1 - p_Y)\rho + p_Y Y\rho Y \rightarrow \frac{1}{2}(\mathbb{1} + \langle Y \rangle Y)$$

- since $Y = iZX$, we can think of this as being the same as a ZX error

A few more common error models: amplitude damping

The operators we apply to the density matrix do not need to be unitary!

For instance we can take a lowering operator defined as

$$L = |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

This does a bit-flip only if we start from the $|1\rangle$ state. It is a useful model to describe **spontaneous decay** when $|1\rangle$ has higher energy than $|0\rangle$.

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Amplitude damping error channel

$$\rho_A = \mathcal{E}_{ad}[\rho] = D_A \rho D_A^\dagger + p_A L \rho L^\dagger \rightarrow |0\rangle\langle 0|$$

Where the diagonal operator D_A can be defined as follows

$$D_A = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p_A} \end{pmatrix} \quad \text{so that} \quad \mathbb{1} = D_A^\dagger D_A + L^\dagger L$$

- important physical mechanism in many devices $T_1 \approx -\log(p_A)$

A few more common error models: depolarizing

- with probability p_D we perform a Pauli transformation according to
 - with probability $1/3$ we perform a Pauli X transformation
 - with probability $1/3$ we perform a Pauli Y transformation
 - with probability $1/3$ we perform a Pauli Z transformation
- otherwise, with probability $1 - p_D$, we do nothing

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Depolarizing error channel

$$\rho_D = \mathcal{E}_{dep}[\rho] = (1 - p_D)\rho + \frac{p_D}{3}X\rho X + \frac{p_D}{3}Y\rho Y + \frac{p_D}{3}Z\rho Z \rightarrow \frac{1}{2}\mathbb{1}$$

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This is the first example of something called **Pauli twirling**

$$\begin{aligned}\rho_{TW} &= \frac{1}{4} \sum_{k=0,1,2,3} \sigma_k \rho \sigma_k \\ &= \frac{1}{8} (\mathbb{1} + \langle X \rangle X + \langle Y \rangle Y + \langle Z \rangle Z) + \frac{1}{8} (\mathbb{1} + \langle X \rangle X - \langle Y \rangle Y - \langle Z \rangle Z) \\ &\quad + \frac{1}{8} (\mathbb{1} - \langle X \rangle X + \langle Y \rangle Y - \langle Z \rangle Z) + \frac{1}{8} (\mathbb{1} - \langle X \rangle X - \langle Y \rangle Y + \langle Z \rangle Z)\end{aligned}$$

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This is everyone favourite error channel for several reasons

- it expresses the intuition behind decoherence: loss of information

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- it can be generalized straightforwardly to n qubit systems
 - local and independent: $\mathcal{E}_{dep}^{1\otimes 2}[\rho] = \mathcal{E}_{dep}^2[\mathcal{E}_{dep}^1[\rho]]$
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Unfortunately for everyone it does not exist in the real world...

Error mitigation with zero-noise extrapolation (ZNE)

In general we can try to quantify the noise level using an infidelity ϵ

$$\mathcal{E}_{noise}[\rho] = (1 - \epsilon)\rho + \epsilon\rho_{error}$$

For small noise levels we can expand measured expectation values as

$$\langle P \rangle(\epsilon) = \text{Tr}[P\mathcal{E}_{noise}[\rho]] = \mathcal{P}_0 + \epsilon\mathcal{P}_1 + \frac{\epsilon^2}{2}\mathcal{P}_2 + \dots$$

where $\mathcal{P}_0 = \langle P \rangle(\epsilon = 0)$ is the ideal expectation value with no noise.

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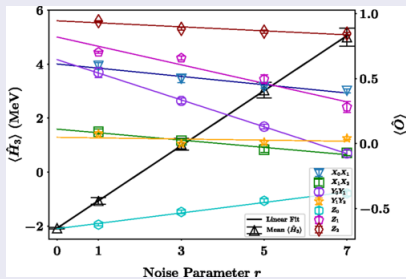
Zero noise extrapolation

Estimate $\langle P \rangle(\epsilon)$ for different noise levels ϵ and extrapolate to $\epsilon = 0$.
Using two points $\epsilon_2 = r\epsilon_1$ we have

$$\mathcal{P}_0 \approx \langle P \rangle(\epsilon_1) - \frac{\langle P \rangle(\epsilon_1) - \langle P \rangle(\epsilon_2)}{r - 1}$$

Li & Benjamin PRX(2017), Temme, Bravy, Gambetta
PRL(2017), Endo, Benjamin, Li PRX(2018)

picture from Dumitrescu et al. PRL(2018)



Error mitigation with zero-noise extrapolation (ZNE)

For a recent review see arXiv:2005.10921

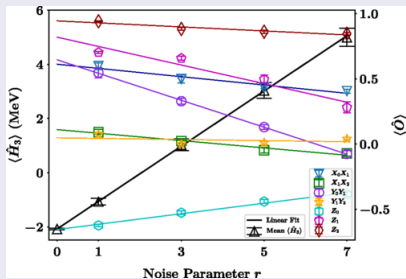
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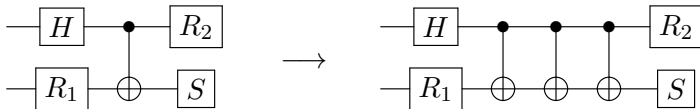
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picture from Dumitrescu et al. PRL(2018)



How do we actually extract $\langle P \rangle(\epsilon)$ at different noise levels? A few ways:

- since noise level tends to grow with time, we can make gates slower
- we can add redundant operations that boost the noise strength



Error mitigation with zero-noise extrapolation (ZNE)

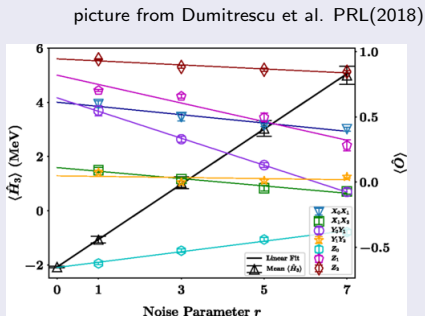
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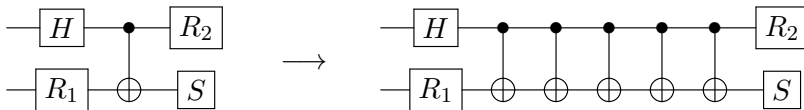
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Observable rescaling/noise learning/self mitigation

Urbanek et al. PRL(2021), Rahman et al. PRD(2022), Farrell et al. PRD(2023),...

Remember the effect of depolarizing noise on Pauli expectation values

$$\langle P \rangle_D = \text{Tr}[P \mathcal{E}_{dep}[\rho]] = (1 - p_D) \langle P \rangle + \frac{p_D}{2} \text{Tr}[P] = (1 - p_D) \langle P \rangle$$

If depolarizing noise was real and we knew the probability p_D then simply

$$\langle P \rangle = \langle P \rangle_D / (1 - p_D)$$

The idea is to learn p_D from a **calibration circuit** and use it as above.

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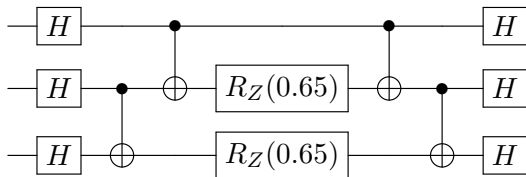
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- need a circuit with similar noise but with known output
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Acts non-trivially on the state $|010\rangle$ but is the identity operation on the state $|000\rangle$ instead

Observable rescaling/noise learning/self mitigation II

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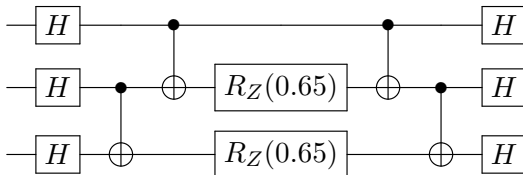
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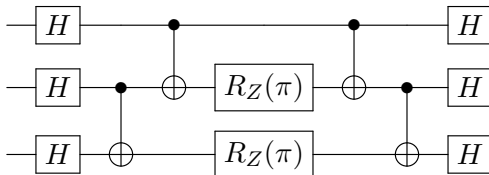
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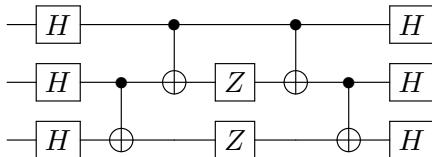
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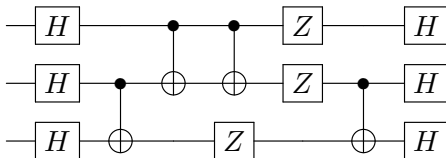
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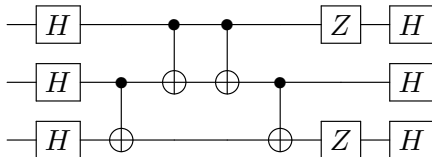
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Observable rescaling/noise learning/self mitigation III

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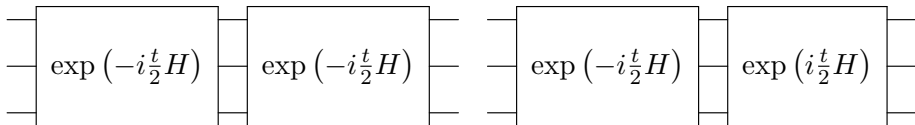
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Observable rescaling/noise learning/self mitigation IV

Urbanek et al. PRL(2021), Rahman et al. PRD(2022), Farrell et al. PRD(2023),...

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In all of these cases we can then do the mitigation as follows

$$\langle P \rangle = \frac{\langle P \rangle_D}{1 - p_D} \approx \langle P \rangle_D \frac{\langle P \rangle_{cal}}{\langle P \rangle_{cal,D}} \quad \left\{ \begin{array}{ll} \langle P \rangle_{cal} & \text{we know without running} \\ \langle P \rangle_{cal,D} & \text{we get from calibration run} \\ \langle P \rangle_D & \text{we get from production run} \end{array} \right.$$

This allows for p_D to be different for different observables.

Observable rescaling/noise learning/self mitigation IV

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In all these cases p_D can be different for different observables.

- need conserved Pauli observable for our circuit and know initial value

$$\text{Tr}[P_{cons} \mathcal{E}_{dep}[\rho]] = (1 - p_D) \langle P_{cons} \rangle = (1 - p_D) \text{Tr}[P_{cons} \rho_0]$$

Notebook time II

We can now take a look at “ErrorMitigation-Gates.ipynb”

-

How do we make the noise into something we like?

Imagine we model a noisy version $\tilde{V}$ of a gate V as the composition

$$\text{---} \boxed{\tilde{V}} \text{---} = \text{---} \boxed{\mathcal{E}} \text{---} \boxed{V} \text{---}$$

We now take a set of unitary gates $\mathcal{U}$ and consider the average

$$\text{---} \boxed{V} \text{---} = \frac{1}{|\mathcal{U}|} \sum_{U \in \mathcal{U}} \text{---} \boxed{U} \text{---} \boxed{V} \text{---} \boxed{U_V} \text{---}$$

where equality is ensured by choosing $U_V V U \equiv V$.

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$$\text{---} \boxed{V} \text{---} = \frac{1}{|\mathcal{U}|} \sum_{U \in \mathcal{U}} \text{---} \boxed{U} \text{---} \boxed{V} \text{---} \boxed{U_V} \text{---}$$

where equality is ensured by choosing $U_V V U \equiv V$. But then with noise

$$\begin{aligned} \frac{1}{|\mathcal{U}|} \sum_{U \in \mathcal{U}} \text{---} \boxed{U} \text{---} \boxed{\tilde{V}} \text{---} \boxed{U_V} \text{---} &= \frac{1}{|\mathcal{U}|} \sum_{U \in \mathcal{U}} \text{---} \boxed{U} \text{---} \boxed{\mathcal{E}} \text{---} \boxed{V} \text{---} \boxed{U_V} \text{---} \\ &= \frac{1}{|\mathcal{U}|} \sum_{U \in \mathcal{U}} \text{---} \boxed{U} \text{---} \boxed{\mathcal{E}} \text{---} \boxed{U^{-1}} \text{---} \boxed{U} \text{---} \boxed{V} \text{---} \boxed{U_V} \text{---} \\ &= \frac{1}{|\mathcal{U}|} \sum_{U \in \mathcal{U}} \text{---} \boxed{U} \text{---} \boxed{\mathcal{E}} \text{---} \boxed{U^{-1}} \text{---} \boxed{V} \text{---} = \text{---} \boxed{\mathcal{E}_{TW}} \text{---} \boxed{V} \text{---} \end{aligned}$$

The process of turning the channel $\mathcal{E}$ into $\mathcal{E}_{TW}$ is called **twirling**

How does twirling help us?

We can turn any channel into a stochastic Pauli channel by twirling

$$\Lambda_{PTW}[\rho] = \frac{1}{2^n} \sum_{P \in \mathcal{P}_n} P \Lambda[P \rho P] P = \sum_{P \in \mathcal{P}_n} c_P P \rho P ,$$

for some Λ -dependent probability distribution c_P with $\sum_P c_P = 1$.

Even though depolarizing noise does not exist in reality, we can always turn any noise in our device into have stochastic Pauli noise

This is particularly efficient if we apply this to *CNOT* or *CZ* gates since

- two qubit gates are the dominant noise source on current devices
- *CNOT* and *CZ* gates are **Clifford operations** (map Pauli into Pauli)

